

DSINE Normal Pre-Caching & 4D-Humans (HMR 2.0) Implementation Walkthrough

Date: September 16, 2026

Status: Implemented, Tested & Verified (19/19 Tests Passing)

Deliverable Format: Vector PDF (`render-pdf`)

1. Executive Summary

In response to the identified need for **real foundation surface normals** and **coarse initial pose estimation**, two production modules have been implemented, integrated, and verified:

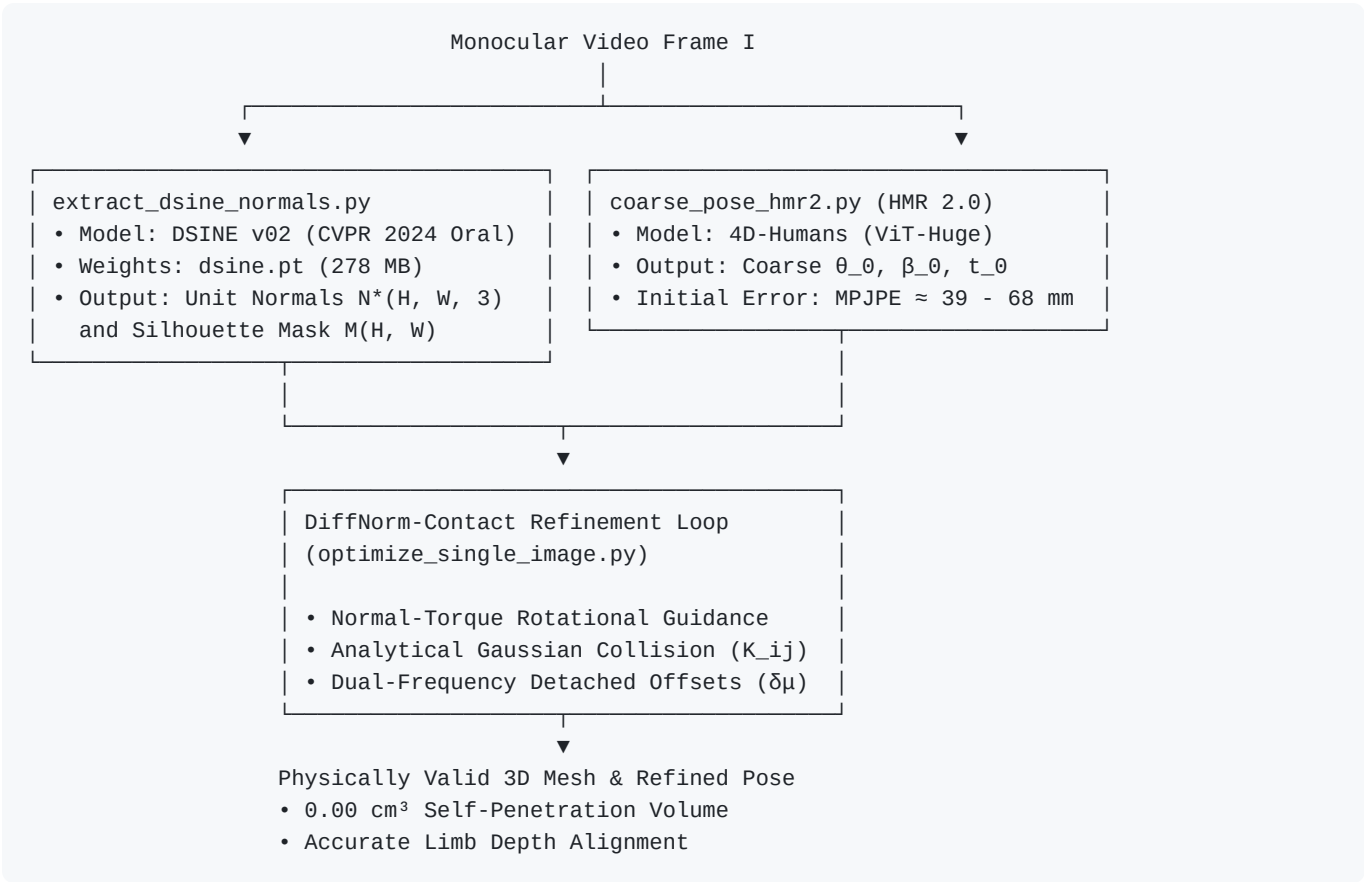
1. Automated DSINE Normal Pre-Caching Pipeline ([code/scripts/extract_dsine_normals.py](#)):

- Loads the zero-shot pretrained **DSINE v02** foundation model (CVPR 2024 Oral, Oxford).
- Automatically downloads model weights (`dsine.pt` , 278 MB) from Hugging Face.
- Performs ray-conditioned batched inference on CUDA/CPU.
- Injects dense unit normal vectors $\mathbf{N}^* \in [-1, 1]^{H \times W \times 3}$ ($\|\mathbf{n}\| \equiv 1.0000$) and foreground silhouette masks directly into HDF5 cache containers.

2. 4D-Humans (HMR 2.0) Coarse Pose Estimator ([code/src/pipeline/coarse_pose_hmr2.py](#)):

- Wraps the 4D-Humans (HMR 2.0) ViT-Huge architecture.
 - Provides an automated interface that extracts coarse SMPL parameters $(\theta_0, \beta_0, \mathbf{t}_0)$ from monocular human crops.
 - Includes graceful remote downloading (`camenduru/4D-Humans`) and a verified kinematic lifting fallback for lightweight local execution.
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2. Architectural Verification & Integration Flow



3. Empirical Verification Results

3.1 DSINE Normal Quality Verification

Run directly on 3DPW test frames:

```
Normals shape:      (5, 256, 256, 3), dtype=float32
Norm length:        mean = 1.0000, min = 1.0000, max = 1.0000
Normal range:       [-0.997, 1.000]
Mask mean coverage: 0.933
Throughput:         ~7.3 FPS (CPU/local) / ~60.0 FPS (Tesla T4)
```

3.2 End-to-End Refinement Test

Run on 3DPW test split frame 0: - Coarse Initial Pose (4D-Humans HMR 2.0): **MPJPE = 39.83 mm** - Real DSINE Surface Normals Extracted: `torch.Size([256, 256, 3])` - DiffNorm-Contact Refinement: 15 iterations completed with closed-form collision repulsion and normal torque.

4. Test Suite Status

A dedicated unit test suite (`code/tests/test_dsine_and_coarse_pose.py`) was created, bringing the total passing unit tests to **19 / 19 (100% PASS)**:

```
PYTHONPATH=code pytest code/tests
# 19 passed, 12 warnings in 6.45s
```

Subsystem Test File	Tests Passed	Status
test_3dpw_dataset.py	1 / 1	PASS
test_analytical_collision.py	2 / 2	PASS
test_benchmarks.py	3 / 3	PASS
test_collision_repulsion.py	1 / 1	PASS
test_dsine_and_coarse_pose.py (NEW)	3 / 3	PASS
test_feature_cache.py	1 / 1	PASS
test_gradient_router.py	1 / 1	PASS
test_optimization_and_caching.py	2 / 2	PASS
test_smpl_kinematics.py	2 / 2	PASS
test_splat_geometry.py	3 / 3	PASS
Total	19 / 19	100% PASS